Dang Nguyen

Email: dangnguyen@uchicago.edu | Website: dangng2004.github.io/ | Google Scholar | Phone: +1 (773) 936-0857

Research Interests

- **AI for science evaluation**: develop AI that can scalably judge science quality to accelerate scientific discovery.
- **AI values**: study the judgment and preferences of AI models and how they can influence humans systems.
- **Alignment**: create methods to align AI models to human values to ensure AI safety.

Education

2024 - Exp. 2029 **Computer Science Ph.D.**, The University of Chicago.
Advisor: Prof. Chenhao Tan

2020 - 2024 **Computer Science B.S. & Mathematics B.A.**, The University of Chicago.
Specialization in Machine Learning.

Preprints

2026 **Benchmarking Agentic Review Systems**. [Link]
Nguyen, D., Hao, W., Elazar, Y., & Tan, C. *Oral presentation (top 10%) and Best Paper Honorable Mention at the AI-Native Academia Workshop at NeurIPS 2026.*

2026 **Moral Mazes in the Era of Large Language Models**. [Link]
Nguyen, D., Fu, H.Y., West, P., Holtzman, A., & Tan, C. *Accepted to the P-AGI Workshop at ICLR 2026.*

2026 **Iterative Finetuning is Mostly Idempotent**. [Link]
Roe, Z.*, Sanderson, J.*, **Nguyen, D.***, Huang, J.*, Nief, T., Shrivastava, A., Tan, C., & Holtzman, A. *Under review.*

Selected Publications

2025 **On the Effectiveness and Generalization of Race Representations for Debiasing High-Stakes Decisions**. [Link]
Nguyen, D., & Tan, C. In Conference on Language Modeling (COLM). *Top 1% of papers in the review process. Best Poster Honorable Mention at Midwest Machine Learning Symposium 2025.*

2025 **GPT-4V Cannot Generate Radiology Reports Yet**. [Link]
Jiang, Y.*, Chen, C.*, **Nguyen, D.**, Mervak, B., & Tan, C. In Findings of the Association for Computational Linguistics (NAACL). *Lightning talk at the Toyota Technological Institute at Chicago (TTIC) Multimodal AI Workshop 2024.*

2023 **Pragmatic Radiology Report Generation**. [Link]
Nguyen, D., Chen, C., He, H., & Tan, C. In Proceedings of Machine Learning for Health (ML4H). *Example paper for ML4H 2024.*

Research Experience

2024 - Present **Chicago Human+AI Lab**, Graduate Researcher.
Advisor: Prof. Chenhao Tan

- Benchmarking Agentic Review Systems (OpenAIReview).
  - Built the web application for OpenAIReview, an open-source LLM paper review system hosted at openaireview.org.
  - Benchmarked 3 review systems across 6 LLMs on detecting injected errors in 320 papers.
- On the Effectiveness and Generalization of Race Representations for Debiasing High-Stakes Decisions.
  - Found causal race representations in Gemma and LLaMA with up to 90% accuracy using Distributed Alignment Search.
  - Reduced racial bias in admissions and hiring decisions by up to 57% with representation-based interventions.

2025 - 2026 **Communication and Intelligence**, Graduate Researcher.
Advisor: Prof. Ari Holtzman, Prof. Chenhao Tan

- Moral Mazes in the Era of Large Language Models.
  - Built HR Simulator, a game benchmarking human and AI corporate communication, with 200+ players.
  - Analyzed 600+ human and LLM-written emails across 5 writer models and 6 LLM judges.
- Iterative Finetuning is Mostly Idempotent.
  - Finetuned 6 Qwen and LLaMA models on their own outputs via the Tinker API to test persona and belief amplification across 6 personas.

2023 - 2024 **Chicago Human+AI Lab**, Undergraduate Researcher.
Advisor: Prof. Chenhao Tan

- Pragmatic Radiology Report Generation.
  - Reduced hallucinations by 69.5% over the previous state of the art by cleaning training reports with Flan-T5.
  - Built a ResNet + LLaMA 7B model outperforming the previous state of the art by up to 52.9% on clinical metrics.

2020 - 2022 **Machine Learning for Systems Research (UCARE Lab)**, Undergraduate Researcher.
Advisor: Prof. Haryadi Gunawi

- Developed statistical and ML-based methods (based on K-means, DBSCAN, KL divergence) for detecting anomalous hard disk drives at scale.

Industry Experience

Summer 2026 **Atoms**, Machine Learning PhD Intern.

- Developed an LLM-assisted framework for non-experts to achieve expert-level annotation of complex human–AI conversation data.
- Used the framework to create a gold dataset of 200 conversations and an LLM labeler achieving over 0.8 held-out F1.
- Developed a taxonomy of AI agent capability gaps with 10 top-level categories.

Summer 2022 **NetApp**, Research Intern.

- Developed ML algorithms for slow disk detection, improving the true positive rate from 10% to 59% over an existing heuristic detector.

Teaching Experience

Spring 2026 **Teaching Assistant**, CMSC 25750: Large Language Models, University of Chicago.
Designed homework assignments and mentored 25 student teams on final projects.

Winter 2026 **Teaching Assistant**, CMSC 25700: Natural Language Processing, University of Chicago.
Designed homework assignments and mentored 9 student teams on final projects.

Winter 2025 **Teaching Assistant**, CMSC 25700: Natural Language Processing, University of Chicago.
Designed homework assignments and mentored 7 student teams on final projects. 3 advised teams ranked in the top 6 out of 26; one won the Best Final Report award.

Service

2026 **Reviewer**, Conference on Language Modeling (COLM).

2026 **Reviewer**, Association for Computational Linguistics (ACL) System Demonstrations.

2025 **Reviewer**, Conference on Language Modeling (COLM).

2025 **Reviewer**, International Conference on Learning Representations (ICLR).

2024 **Reviewer**, Machine Learning for Health (ML4H).